

Written Justification

System: Vaccination Cohort & Dose Scheduling System

Architecture Selection

We chose **Layered Architecture** for the **Vaccination Cohort & Dose Scheduling System**.

Architectural Choice

The selected architectural style is **Layered Architecture**, consisting of a Presentation Layer, Business Logic Layer, and Data Layer.

Reason 1

Layered Architecture provides clear separation between user interaction, vaccination-related business operations, and data storage. This makes functions such as citizen registration, dose scheduling, and vaccination record management easier to organize.

Reason 2

The system has several related functions, including checking the minimum interval between vaccine doses, booking vaccination slots, recording administered doses, and generating certificates. Separating these functions into appropriate layers makes the system easier to maintain and modify.

Security Advantage

The layered structure allows access to vaccination records to be controlled through the Business Logic Layer. Users do not directly access the database, allowing authentication and authorization to be applied before sensitive vaccination information is accessed.

Performance Benefit

The Business Logic Layer can efficiently process operations such as dose eligibility checking and slot scheduling, while the Data Layer retrieves only the required vaccination and appointment information from the database. This helps reduce unnecessary data access and improves system responsiveness.